

HMMT 2014
Saturday 22 February 2014
HMIC

1. [20] Consider a regular n -gon with $n > 3$, and call a line *acceptable* if it passes through the interior of this n -gon. Draw m different acceptable lines, so that the n -gon is divided into several smaller polygons.
 - (a) Prove that there exists an m , depending only on n , such that any collection of m acceptable lines results in one of the smaller polygons having 3 or 4 sides.
 - (b) Find the smallest possible m which guarantees that at least one of the smaller polygons will have 3 or 4 sides.

Answer: N/A We will prove that if $m \geq n - 4$, then there is guaranteed to be a smaller polygon with 3 or 4 sides, while if $m \leq n - 5$, there might not be a polygon with 3 or 4 sides. This will solve both parts of the problem.

Given a configuration of lines, let $P_1, \dots, P_k$ be all of the resulting smaller polygons. Let $E(P_i)$ be the number of edges in polygon P_i , and let $E = E(P_1) + \dots + E(P_k)$. First, note that whenever a new polygon is formed, it must have been because a larger polygon was split into two smaller polygons by a line passing through it. When this happens, k increases by 1 and E increases by at most 4 (it might be less than 4 if the line passes through vertices of the larger polygon). Therefore, if adding an acceptable line increases the number of polygons by a , then E increases by at most $4a$.

Now, assume $m \geq n - 4$. At the beginning, we have $k = 1$ and $E = n$. If the number of polygons at the end is $p + 1$, then $E \leq n + 4p$, so the average number of edges per polygon is less than or equal to $\frac{n+4p}{1+p}$. Now, note that each acceptable line introduces at least one new polygon, so $p \geq m \geq n - 4$. Also, note that as p increases, $\frac{n+4p}{1+p}$ strictly decreases, so it is maximized at $p = n - 4$, where $\frac{n+4p}{1+p} = \frac{5n-16}{n-3} = 5 - \frac{1}{n-3} < 5$. Therefore, the average number of edges per polygon is less than 5, so there must exist a polygon with either 3 or 4 edges, as desired.

We will now show that if $m \leq n - 5$, then we can draw m acceptable lines in a regular n -gon $A_1A_2 \dots A_n$ such that there are no polygons with 3 or 4 sides. Let M_1 be the midpoint of A_3A_4 , M_2 be the midpoint of A_4A_5 , ..., M_{n-5} be the midpoint of $A_{n-3}A_{n-4}$. Let the m acceptable lines be $A_1M_1, A_1M_2, \dots, A_1M_m$. We can see that all resulting polygons have 5 sides or more, so we are done.

2. [25] 2014 triangles have non-overlapping interiors contained in a circle of radius 1. What is the largest possible value of the sum of their areas?

Answer: N/A This problem turned out to be much trickier than we expected. We have yet to see a complete solution, but let us know if you find one!

Comment. We apologize for the oversight on our part. Our test-solvers essentially all misread the problem to contain the additional assumption that all triangle vertices lie on the circumference (which is not the case).

3. [30] Fix positive integers m and n . Suppose that $a_1, a_2, \dots, a_m$ are reals, and that pairwise distinct vectors $v_1, \dots, v_m \in \mathbb{R}^n$ satisfy

$$\sum_{j \neq i} a_j \frac{v_j - v_i}{\|v_j - v_i\|^3} = 0$$

for $i = 1, 2, \dots, m$.

Prove that

$$\sum_{1 \leq i < j \leq m} \frac{a_i a_j}{\|v_j - v_i\|} = 0.$$

Answer: N/A Since $v_i \cdot (v_j - v_i) + v_j \cdot (v_i - v_j) = -\|v_j - v_i\|^2$ for any $1 \leq i < j \leq m$, we have

$$0 = \sum_{i=1}^m a_i v_i \cdot 0 = \sum_{i=1}^m a_i v_i \sum_{j \neq i} a_j \frac{v_j - v_i}{\|v_j - v_i\|^3} = - \sum_{1 \leq i < j \leq m} \frac{a_i a_j}{\|v_j - v_i\|},$$

as desired.

Alternative Solution. Fix $a_1, \dots, a_m$, and define $f : (\mathbb{R}^n)^m \rightarrow \mathbb{R}$ by

$$f(x_1, \dots, x_m) = \sum_{1 \leq i < j \leq m} \frac{a_i a_j}{\|x_j - x_i\|}.$$

Now if we view $f(x_1, \dots, x_m)$ individually as a function in x_k (k fixed), then the problem condition for $i = k$ says precisely that the gradient of f with respect to x_k (but of course, multiplied by a_k) evaluated at $(v_1, \dots, v_m)$ is 0.

Finally, the multivariate chain rule shows that $F(t) := f(tv_1, \dots, tv_m) = \frac{1}{t} f(v_1, \dots, v_m)$ must satisfy $F'(1) = 0$. Yet $F'(1) = -\frac{1}{1^2} f(v_1, \dots, v_m)$, so we have $f(v_1, \dots, v_m) = 0$, as desired.

Comment. This is (sort of) an energy minimization problem from physics (more precisely, minimizing electrostatic or gravitational potential).

Comment. It would be interesting to determine all differentiable $A : \mathbb{R}^+ \rightarrow \mathbb{R}$ such that the problem still holds when the terms are replaced by $\frac{v_j - v_i}{\|v_j - v_i\|} A'(\|v_j - v_i\|)$ and $A(\|v_j - v_i\|)$. (Here $A(t) = 1/t$, which has a nice physical interpretation, but that's not important.) The obvious examples are $A(t) = ct^s$ for constant c and real s (where both solutions easily carry through). But there may be others where instead of defining $F(t)$ through the scaling transformation $(v_1, \dots, v_m) \rightarrow (tv_1, \dots, tv_m)$, there's a subtler translation.

4. [35] Let ω be a root of unity and f be a polynomial with integer coefficients. Show that if $|f(\omega)| = 1$, then $f(\omega)$ is also a root of unity.

Answer: N/A Suppose ω is a primitive n th root, so that $\Phi_n(x)$ is the minimal polynomial of ω (over $\mathbb{Q}$). Thus $f(\omega)f(\omega^{-1}) = 1$ implies $\Phi_n(x) \mid f(x)f(x^{-1}) - 1$ (divisibility as Laurent polynomials). Hence $|f(\omega^k)| = 1$ whenever $\gcd(k, n) = 1$, and $\prod_{\gcd(k, n)=1} (t - f(\omega^k))$ is a monic integer-coefficient polynomial with all roots within the unit disk, so its roots must all be roots of unity (since 0 is clearly not a root), and we're done.

Comment. The result used at the end (a monic integer-coefficient polynomial with all roots within the unit disk has every root either 0 or a root of unity) is classical. See for instance the discussion here or here. This result is sometimes called Kronecker's theorem, and appears, for instance, in *Problems from the Book*.

Comment. Note that the proof works as long as $f \in \mathbb{Q}[x]$ and $f(\omega)$ is an algebraic integer. *Once is it known that $f(\omega)$ is a root of unity*, more can be said about the order m of $f(\omega)$. If ω is a primitive n th root of unity, then the conjugates of $f(\omega)$ lie among the $\leq \phi(n)$ numbers in $\{f(\omega^k) : \gcd(k, n) = 1\}$. In particular, $\phi(m) \leq \phi(n)$, which (e.g. through a naive estimate using $p - 1 \geq \sqrt{p}$ for $p \geq 3$ to get $\phi(m) \geq \sqrt{m/2}$) shows that $m \leq 2\phi(n)^2$ (for instance), so there exists $g \in \mathbb{Q}[x]$ with $g(\omega)$ a root of unity of maximal order N , which then must generate the unit group of $\mathbb{Q}(\omega)$ (or else we could find a larger order). In particular, since ω lies in this group, $n \mid N$. But $\phi(N) \leq \phi(n)$ by our earlier argument, so either $N = n$ or $N = 2n$ if n is odd. Either way, it's clear that $g(\omega) = \pm\omega^k$ for some k , so our original $f(\omega)$ must also be $\pm\omega^j$ for some j .

Alternative Solution. We follow the paper here, but fill in some of the details to make the ideas as accessible as possible. We will use the fact that if A is a finitely generated abelian group, then any subgroup B is also finitely generated, and furthermore, $\text{rank}(B) \leq \text{rank}(A)$. (For instance, see Theorem 14.6.5 in Artin's *Algebra*, 3rd ed. for a proof of finite generation, which for our particular case of modules over a PID (namely, $\mathbb{Z}$) can be done in parallel to the $\text{rank}(B) \leq \text{rank}(A)$ proof—the proof is very similar to Gaussian elimination.)

Let $K = \mathbb{Q}(\omega)$, V the set of algebraic integers in K of modulus 1 (which must be units of K), $U \supseteq V$ the set of units in K , $R = U \cap \mathbb{R}$ the set of real units in K (which, importantly, is also the set of units in $K \cap \mathbb{R}$), and $L = K \cap \mathbb{R}$ denote the maximal real subfield of K .

The key is that $\text{rank}(V) = \text{rank}(U) - \text{rank}(R)$. To prove this, we note that for $u \in U$, $\bar{u} \in U$ as well (since $\bar{K} = K$), so $u\bar{u} = |u|^2 \in R$ and $u/\bar{u} \in V$, which gives $u^2 = (u\bar{u})(u/\bar{u}) \in RV$. Thus $S := \{u^2 : u \in U\} \subseteq RV \subseteq U$. But U is a finitely generated abelian group by Dirichlet's unit theorem, so S, RV are as well, and furthermore $\text{rank}(S) \leq \text{rank}(RV) \leq \text{rank}(U)$. Now take multiplicatively independent $u_1, \dots, u_{\text{rank}(U)} \in U$; then clearly $u_1^2, \dots, u_{\text{rank}(U)}^2 \in S$ are multiplicatively independent as well, so $\text{rank}(U) \leq \text{rank}(S)$, whence $\text{rank}(RV) = \text{rank}(U)$. Of course, $R \cap V = \{\pm 1\}$, so $\text{rank}(RV) = \text{rank}(R) + \text{rank}(V) - \text{rank}(R \cap V) = \text{rank}(R) + \text{rank}(V)$ (the proof here for $\mathbb{Z}$ -modules is essentially the same as that for $\mathbb{Q}$ -vector spaces), finishing the proof.

V contains only roots of unity if and only if $\text{rank}(V) = 0$ (no free elements), so we must prove $\text{rank}(U) = \text{rank}(R)$. Let r, s denote the number of real and complex embeddings of a number field, so $[K : \mathbb{Q}] = r(K) + 2s(K)$, $[L : \mathbb{Q}] = r(L) + 2s(L)$, and thus $r(K) + 2s(K) = [K : L](r(L) + 2s(L))$. By Dirichlet's unit theorem, $\text{rank}(U) = r(K) + s(K) - 1$ and $\text{rank}(R) = r(L) + s(L) - 1$.

If $K = L$ (equivalently, K is totally real), we're trivially done. Otherwise, if $[K : L] > 1$, it's easy to check that we have $\text{rank}(U) = \text{rank}(R)$ if and only if $[K : L] = 2$ and $r(K) = s(L) = 0$ (K is totally imaginary and L is totally real)—then $r(L) = s(K)$ automatically holds. This is precisely the condition that K is a CM-field.

(CM-field K just means K is totally imaginary (no real embeddings), and K is a quadratic extension of $K \cap \mathbb{R}$.)

In particular, the cyclotomic field $K = \mathbb{Q}(\omega) = \mathbb{Q}[\omega]$ is a totally imaginary quadratic extension of the totally real field $\mathbb{Q}(\omega + \omega^{-1}) = \mathbb{Q}[\omega + \omega^{-1}]$. Indeed, it is easy to check that the latter is simply $K \cap \mathbb{R}$, and $K = \mathbb{Q}(\omega + \omega^{-1}, \omega - \omega^{-1} = \pm\sqrt{(\omega + \omega^{-1})^2 - 4})$. So we're done.

5. [40] Let n be a positive integer, and let A and B be $n \times n$ matrices with complex entries such that $A^2 = B^2$. Show that there exists an $n \times n$ invertible matrix S with complex entries that satisfies $S(AB - BA) = (BA - AB)S$.

Answer: $\boxed{N/A}$ Let $X = A + B$ and $Y = A - B$, so $XY = BA - AB$ and $YX = AB - BA$. Note that $XY = -YX$.

It suffices (actually is equivalent) to show that $AB - BA$ and $BA - AB$ have the same Jordan forms. In other words, we need to show that for any complex number λ , the Jordan λ -block decompositions of $AB - BA$ and $BA - AB$ are the same.

However, the λ -decomposition of a matrix M is uniquely determined by the infinite (eventually constant) sequence $(\dim \ker(M - \lambda I)^1, \dim \ker(M - \lambda I)^2, \dots)$, so we only have to show that $\dim \ker(XY - \lambda I)^k = \dim \ker(YX - \lambda I)^k$ for all positive integers k and complex numbers λ .

For $\lambda = 0$, this follows trivially from $XY = -YX$: we have $(XY)^k v = 0$ if and only if $(YX)^k v = 0$.

Now suppose $\lambda \neq 0$, and define the (matrix) polynomial $p(T) = (T - \lambda I)^k$. Let $V_1 = \{v \in \mathbb{C}^n : p(XY)v = 0\}$, and $V_2 = \{v \in \mathbb{C}^n : p(YX)v = 0\}$. Note that if $v \in V_1$, then $0 = [Yp(XY)]v = [p(YX)Y]v = p(YX)[Yv]$, so $Yv \in V_2$. Furthermore, if $v \neq 0$, then $Yv \neq 0$, or else $p(XY)v = p(0)v = (-\lambda I)^k v \neq 0$ (since $(XY)^j v = 0$ for all $j \geq 1$), contradiction. Viewing Y (more precisely, $v \mapsto Yv$) as a linear map from V_1 to V_2 , we have $\dim V_1 = \dim YV_1 + \dim \ker Y = \dim YV_1 \leq \dim V_2$ (alternatively, if $v_1, \dots, v_r$ is a basis of V_1 , then $Yv_1, \dots, Yv_r$ are linearly independent in V_2). By symmetry, $\dim V_1 = \dim V_2$, and we're done.

Comment. If A, B are additionally real, then we can also choose S to be *real* (not just complex). Indeed, note that if $S = P + Qi$ for real P, Q , then since A, B are real, it suffices to find $c \in \mathbb{R}$ such that $\det(P + cQ) \neq 0$. However, $\det(P + xQ)$ is a polynomial in x that is not identically zero (it is nonzero for $x = i$), so it has finitely many (complex, and thus real) roots. Thus the desired c clearly exists.

Comment. Note that $A(AB - BA) = (BA - AB)A$, since $A^2B = B^3 = BA^2$. (The same holds for $A \rightarrow B$.) Thus the problem is easy unless all linear combinations of A, B are noninvertible. However, it seems difficult to finish without analyzing generalized eigenvectors as above.

Comment. See this paper by Ross Lippert and Gilbert Strang for a general analysis of the Jordan forms of MN and NM .